

Q 1 a) Robot Navigation

Formulation

- ① State : (row, col) of the robot
- ② Initial State : (0,0), where S is
- ③ Goal State : (4, 6) where G is
- ④ Actions : U, D, L, R
- ⑤ Successor function : each move whose new cell is inside grid (not #)
- ⑥ Cost : 1 / move
- ⑦ Heuristic : Manhattan distance to G
 $\rightarrow$ never overestimates

$\rightarrow$ heuristic values for goal (4,6)

$\rightarrow$ works $h_g = (4 - \text{row}) + (6 - \text{col})$

$h=0$ to $h=10$

10	9	8	7	6	5	4
9	8	7	6	5	4	3
8	7	6	5	4	3	2
7	6	5	4	3	2	1
6	5	4	3	2	1	0

State	U	D	L	R	Successors
(0,0)	-	#	-	free	just R
(2,2)	free	#	free	free	U, L, R
(4,4)	free	-	free	#	U, L

The A* trace path : RRDDRDRRDD

cost : 10

Nodes expanded : 12 (see below @*)

from our run we see

Dijkstra takes

expands

24 nodes

Formulation Courier Delivery → load CSV once with Load Courier Data() & parameters

- ① State: the area the courier is in (name)
- ② Initial State: start area (i.e. hub = Saddar)
- ③ Goal: delivery area
- ④ Actions: drive to neighbouring area
- ⑤ Successors: area with a direct road (i.e. NOT a-1 in Connections.csv)
- ⑥ Cost: km × road type number
- ⑦ Heuristic: straight line distance from heuristics.csv

Road Type Numbers	
M	1.0
S	1.25
N	3.0

main roads are preferred

checking edge costs

	km	type	cost
Saddar → II Chundigar Road	3	M	$3 \times 1.0 = 3.0$
Saddar → PECHS	6	S	$6 \times 1.25 = 7.5$
Liagatbad → Nazimabad	3	S	$3 \times 1.25 = 3.75$
Saddar → Lyari	5	N	$5 \times 3.0 = 15.0$

why are heuristics safe?

→ roads never shorter than straight line, so cost never below
→ Saddar to see below

see undirected values

A* trace: Saddar (Hub) to Gulshan-e-Iqbal.

$f = g + h$	Pop	g	h	f	Frontier (node (g, f))	area	h to Gulshan-e-Iqbal
	Saddar (Hub)	0	7.6	7.6	II Chundigar (3.0, 3), PECHS (7.5, 11.6), Liagatbad (8.75, 13.75), Clifton (8.0, 16.6), Lyari (15.0, 24.2)	Saddar	7.6
	II Chundigar Road	3.0	6.3	9.3	PECHS, Liagatbad, Clifton, Lyari (same (g, f) except for Lyari which is 12)	II Chundigar	6.3
	PECHS	7.5	4.1	11.6	Gulshan (13.5, 13.5), Liagatbad (8.75, 13.75), Clifton, Nazimabad (13.75, 18.25), Lyari	PECHS	4.1
	Gulshan-e-Iqbal	13.5	0	13.5	goal reached	Liagatbad	5.0
						Clifton	8.6
						Lyari	9.2
						Nazimabad	4.5

Path: Saddar → PECHS → Gulshan-e-Iqbal

Cost: ~~15.0~~ ~~13.5~~ 13.5

Nodes visited: 4

Dijkstra on the same trip ($m=0$, so $f=9$)

changed
Pop order: Saddar (0), chundigar (3.0), PECHS (7.5), Clifton (8.0), Liaquatabad (8.75),
Nazimabad (12.5), Gulshan-e-Iqbal (13.5).

→ nodes visited: 7

what is the road type change? Saddar → Baldia Town $\mathbb{P}$

route without $\rightarrow$ Saddar → Lyari → Baldia Town, Cost = $5+9=14$

with $\rightarrow$ Saddar → Liaquatabad, Nazimabad, SITE area → Baldia Town, Cost = 28.125

M:1, N:3, S:1.25

better than

→ Lyari to route which now

costs

$$5(3) + 9(3) = 42$$

81(c) Planning → Don't latex document

Robot tests

$(0,0) \rightarrow (4,6)$
 $(0,0) \rightarrow (4,0)$
 $(0,0) \rightarrow (2,4)$
 $(4,0) \rightarrow (0,6)$ } example
5x7

$(0,0) \rightarrow (9,9)$, $(0,0) \rightarrow (0,5)$
 $(2,3) \rightarrow (8,7)$ } open 10x10,
no walls

$(3,1) \rightarrow (3,7)$
 $(0,0) \rightarrow (0,9)$
 $(2,2) \rightarrow (2,4)$ } wall 7x10,
wall in middle

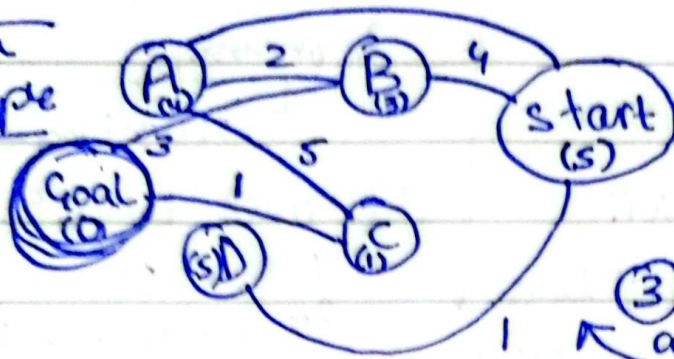
Courier tests

Prough work

① just added costs

② Since h never over estimates, its gonna be equal

Q1(b) Jon example



heuristic

A* trace

from	to below, dis to goal
A	2+3 = 5
B	3
C	1
Start	1+2+3 = 6
D	7

③ add him brackets

$f = g + h \rightarrow$ pop the lowest f (for ties pop whichever was added first)

Pop	g	f	Expanded so far	Frontier (Node(g,f))
S	0	5	1	A(1,5), D(1,6), B(4,7)
A	1	5	2	D(1,6), B(3,6), C(6,7)
D	1	6	3	B(3,6), C(6,7)
B	3	6	4	C(6,7), G(6,6)
G	6	6	5	Goal reached ✓

@ A, B improves from $g=4$ to $g=3$ so old entry removed

Path = $S \rightarrow A \rightarrow B \rightarrow G$

Cost = 6
Nodes expanded = 5

Done

Dijkstra trace. (Same ^{as A*} graph but $h = 0$ so $f = g + 0, f = g$)

Pop	g	Expanded so far	Frontier (node(g))
S	0	1	A(1), D(1), B(4)
A	1	2	D(1), B(3), C(6)
D	1	3	B(3), C(6)
B	3	4	C(6), G(6)
② B (old one with $g=4$)	4	skipped, doesn't count	—
C	6	5	G(6)
G	6	6	Goal Reached

① Goal is checked when a node is popped, not when it's added

② A popped entry whose cost is higher than the best known for that node is skipped.

same path as A* but 6 nodes expanded instead of 5